

Chapter 1

Basic equations of linear elasticity and Principal Stresses and Strain

1.1 Question Bank

Quiz Questions

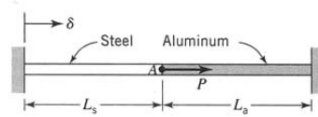
1. The deformation of a bar under its own weight as compared to that when subjected to a direct axial load equal to its own weight will be....
2. The highest stress that a material can withstand for a specified length of time without excessive deformation is called
3. The stretch in a steel rod of circular section having length L subjected to a tensile load P and tapering uniformly from diameter D_1 to D_2 is given as
4. A block of steel is loaded by a tangential force on its top surface while the bottom surface is held rigidly. The deformation of the block is due to.....
5. When tensile stress is applied axially on a circular rod its diameter.....
6. For a linearly elastic, isotropic and homogeneous material, the number of elastic constants required to relate stress and strain is.....
7. The material that exhibits the same elastic properties in all directions at a point is said to be.....
8. What is an incompressible medium, and what is its Poissons ratio?
9. The relation between the stiffness matrix K and the Compliance Matrix C is
10. The Young's modulus of elasticity of a material is 2.5 times its modulus of rigidity. The Poisson's ratio for the material will be:.....
11. The relation between E , G , K and μ are.....
12. Write the Compliance matrix for an Anisotropic Material.
13. For any engineering material, Poisson's ratio limiting values are between.....
14. If Elastic modulus of material is 200 Gpa and Poisson's ratio is 0.29, bulk modulus is....
15. Load divided by initial cross-section area of specimen before deformation is called.....

16. If design load of a material is 20kN and factor of safety is two, then ultimate stress will be.....
17. In a general two dimensional stress system, planes of maximum shear stress are inclined at with principal planes.
18. If a piece of material neither expands nor contracts in volume when subjected to stress, then the Poissons ratio must be.....
19. The phenomenon of progressive extension of the material i.e., strain increasing with time at a constant load is called.....
20. The number of components in a stress tensor defining stress at a point in three dimensions is.....
21. Two examples of plane stress conditions are.....
22. The number of elastic constants for a completely anisotropic elastic material which follows Hooke's law is.....
23. Materials which show direction dependent properties are called.....
24. In terms of Poisson's ratio the ratio of Young's Modulus (E) to Shear Modulus (G) of elastic materials is.....
25. A weight falls on a plunger fitted in container filled with oil thereby producing a pressure of 1.5 N/mm² in the oil. The bulk modulus of oil is 2800 N/mm². Given this situation the volumetric compressive strain produced in the oil will be
26. A cube with a side of length of 1cm is heated uniformly to 1° C above the room temperature and all sides are free to expand. What will be the increase in volume of the cube? Assume thermal expansion is α per ° C
27. what is an inelastic material?
28. Two tapering bars of the same material are subjected to a tensile load of P. The lengths of both the bars are the same. The larger of each of the bars is D. The diameter of the bar A at its smaller end is D/2 and that of the bar B is D/3. What is the ratio of elongation of the bar A to that of the bar B?
29. For any engineering material, poissos ratio limiting values are between
30. If elastic modulus of a material is 200 GPa and poissos ratio is 0.29, then bulk modulus is
31. Load divided by initial cross section are of specimen before deformation is called
32. If a material is subjected to two incremental true strains namely ϵ_1 and ϵ_2 , then the total true strain is
33. The relationship between stress and strain defined in hookes law is independent of
34. The radius of a wire stretched by load is reduced by half the youngs modulus will be

35. What would be the Poisson's ratio of an incompressible medium?
36. If μ is 0.5 the ratio of G to E will be

Main Questions

1. Derive the Generalized Hooke's law in the Compliance form and, Write the compliance matrix.
2. Derive the relationship between the linear elastic properties Young's modulus (E), rigidity modulus (G) and bulk modulus (K)?
3. What different stresses are set-up in a bolt due to initial tightening, while used as a fastener? Name all the stresses in detail.
4. Draw the stress-strain curve for a ductile material and explain salient points on the curve.
5. A steel bar and an aluminum bar are joined end to end and fixed between two rigid walls as shown in Figure. The cross-sectional area of the steel bar is A , and that of the aluminum bar is A_1 . Initially, the two bars are stress free. Derive general expressions for the deflection of point A, the stress in the steel bar, and the stress in the aluminum bar when
 - (a) Load P is applied at point A
 - (b) The left wall is displaced an amount δ to the right.



6. Prove that shear stress τ increases to a maximum value at 45 degrees when a uni-axial stress is applied over an oblique plane.
7. What are principle planes? Explain Principle stresses and strains and what is their significance.
8. Find the principal stresses and the corresponding directions from the stress tensor.

$$\sigma = \begin{bmatrix} 3 & 4 & 6 \\ 4 & 2 & 5 \\ 6 & 5 & 1 \end{bmatrix}$$
9. Derive the expression for axial deformation of a bar with constant cross section. A steel wire 2m long and 3 mm in diameter is extended by 0.75 mm. When a weight W is suspended from the wire. If the same weight is suspended from a brass wire, 2.5 m long and 2 mm in diameter. It is elongated by 4.64 mm. Determine the modulus of elasticity of brass if that of steel be $2 \cdot 10^5 \text{ N/mm}^2$
10. A steel bolt of 20 mm diameter passes centrally through a copper tube of internal diameter 28 mm and external diameter 40 mm. The length of whole assembly is 600 mm. After tight fitting of the assembly, the nut is over tightened by quarter of a turn. What are the stresses introduced in the bolt and tube, if pitch of nut is 2 mm? Take $E_s = 2 \cdot 10^5 \text{ N/mm}^2$ and $E_c = 1.2 \cdot 10^5 \text{ N/mm}^2$.

11. A cylindrical specimen of a titanium alloy having an elastic modulus of 108 GPa and an original diameter of 3.9 mm will experience only elastic deformation when a tensile load of 2000 N is applied. Compute the maximum length of the specimen before deformation if the maximum allowable elongation is 0.42 mm. And list at least two factors that promote transition from ductile to brittle fracture.
12. A composite rod is 1000 mm long, its two ends are 40 mm² and 30 mm² in area and length are 300 mm and 200 mm respectively. The middle portion of the rod is 20 mm² in area and 500 mm long. If the rod is subjected to an axial tensile load of 1000 N, find its total elongation. (E = 200 GPa).
13. Derive the relationship for a deflection of a bar under uniaxial tension. The assembly consists of two 10-mm diameter red brass C83400 copper rods AB and CD, a 15-mm diameter 304 stainless steel rod EF, and a rigid bar G. If P = 5 kN, determine the horizontal displacement of end F of rod EF. (Take E for brass to be 101 GPa and E for steel to be 193 GPa)
14. A bar of length L tapers uniformly from diameter 1.1 D at one end to 0.9 D at the other end. The elongation due to axial pull is computed using mean diameter D. What is the approximate error in computed elongation?
15. A cylindrical specimen of a titanium alloy having an elastic modulus of 108 GPa and an original diameter of 3.9 mm will experience only elastic deformation when a tensile load of 2000 N is applied. Compute the maximum length of the specimen before deformation if the maximum allowable elongation is 0.42 mm.
16. A metallic bar 300×100×40 is subjected to a force of 5kN, 6kN and 4kN along x,y and z directions respectively. Evaluate the change in the volume of the block. Take youngs modulus to be 200GPa and poissons ratio as 0.25.
17. Explain the significance of constitutive equations in elasticity problems.
18. A bar of circular cross section having two different diameters D and 2D, have lengths of L/2 each. Calculate the strain energy if load P=27kN, length L=600mm and diameter D=40mm and E=105GPa.
19. Prove that $\sigma_{x'x'} + \sigma_{y'y'} = \sigma_{xx} + \sigma_{yy}$
20. Prove that in a two dimensional stress system, planes of maximum shear stress are inclined at 45° with the principal planes.